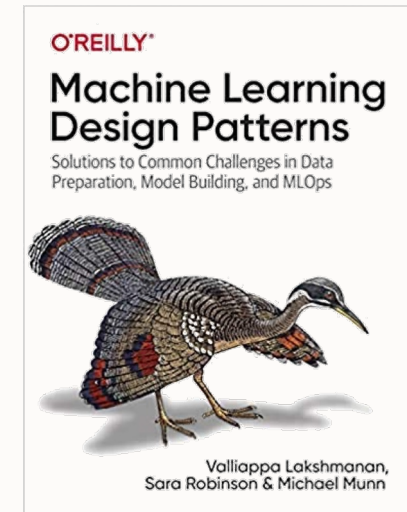




Patrons de conception pour MLOps

Durée 3h

MLOps DP & BP (1)

- Patrons de conception pour la représentation des données :
 1. Hashed Feature
 2. Embedding
 3. Feature Cross
 4. Multimodal Input

MLOps DP & BP (2)

- Patrons de conception pour la représentation des problèmes:
 1. Reframing
 2. Multilabel
 3. Ensemble
 4. Cascade
 5. Neutral Class
 6. Rebalancing

MLOps DP & BP (3)

- Patrons de conception pour l'apprentissage :
 1. Useful overfitting
 2. Checkpoints
 3. Transfer Learning
 4. Distribution Strategy
 5. Hyperparameter Tuning

MLOps DP & BP (4)

- Patrons de conception pour la résilience :
 1. Stateless Serving Function
 2. Batch Serving
 3. Continuous Model Evaluation
 4. Two Phase Predictions
 5. Keyed Predictions

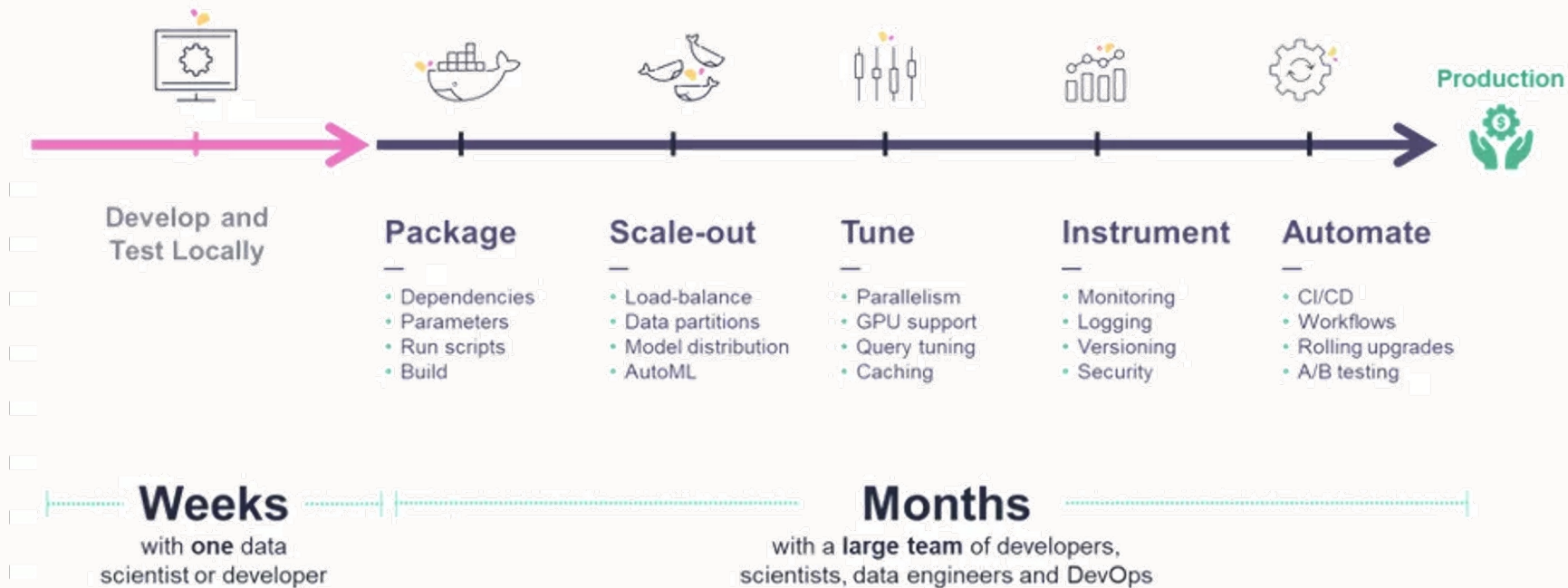
MLOps DP & BP (5)

- Patrons de conception pour la reproductibilité :
 1. Transform
 2. Repeatable Sampling
 3. Bridged Schema
 4. Windowed Inference
 5. Workflow Pipeline
 6. Feature Store
 7. Model Versioning

MLOps DP & BP (6)

- Partons de conception pour une IA responsable :
 1. Heuristic benchmark
 2. Explainable Predictions
 3. Fairness Lens

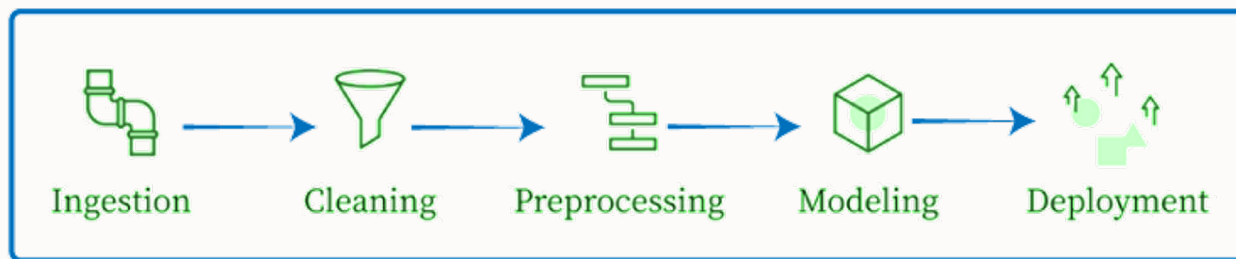
MLOps DP & BP (7)



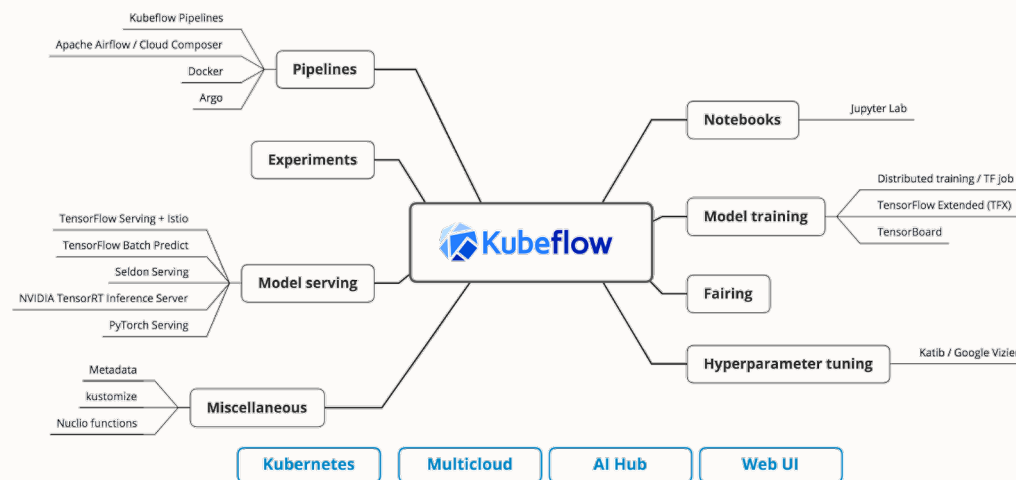
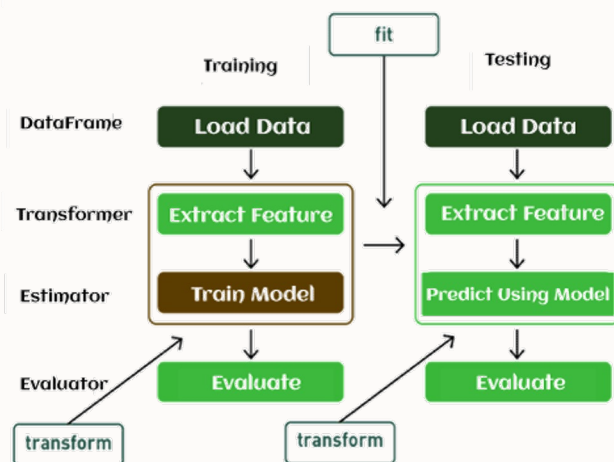
Design Pattern: Workflow Pipeline (1)

MLOps DP & BP (8)

Machine Learning Workflow

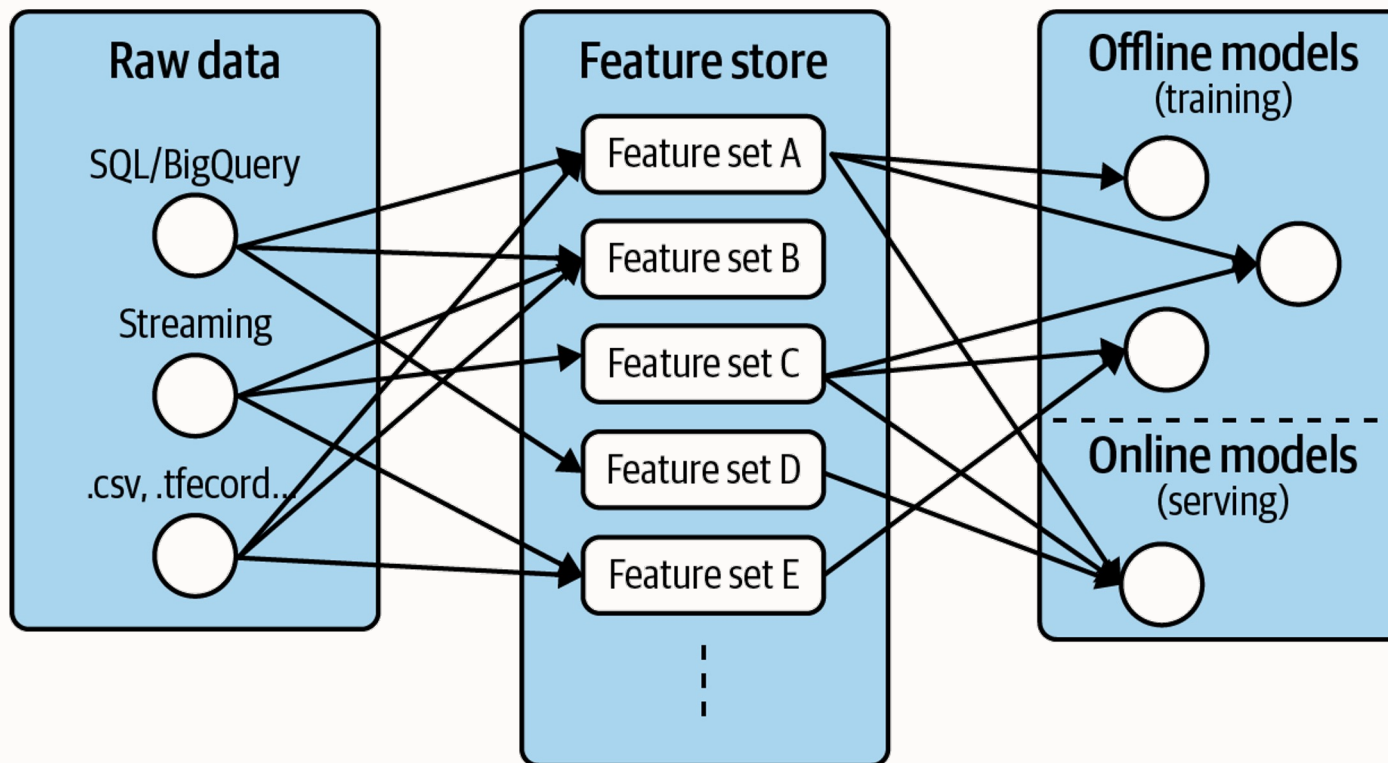


Spark ML Workflow



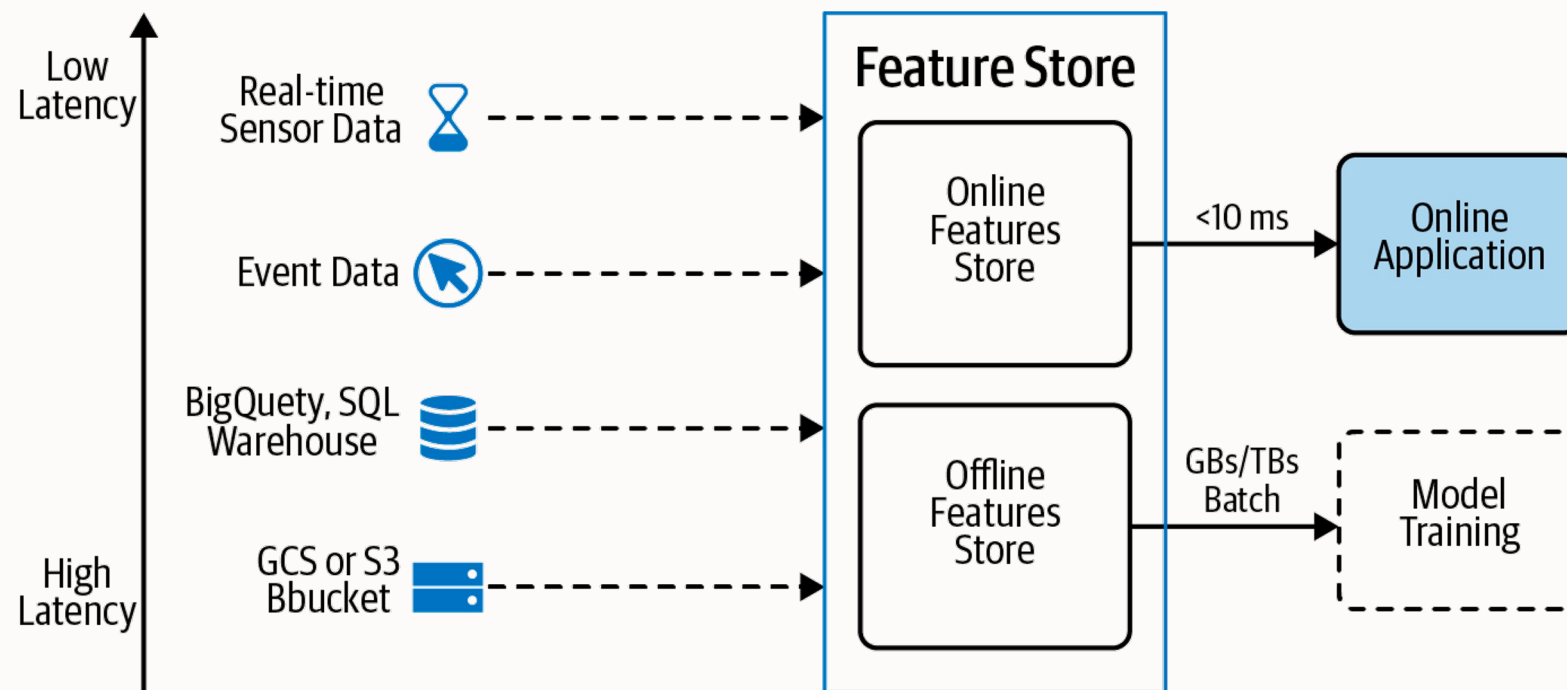
Design Pattern: Workflow Pipeline (2)

MLOps DP & BP (9)



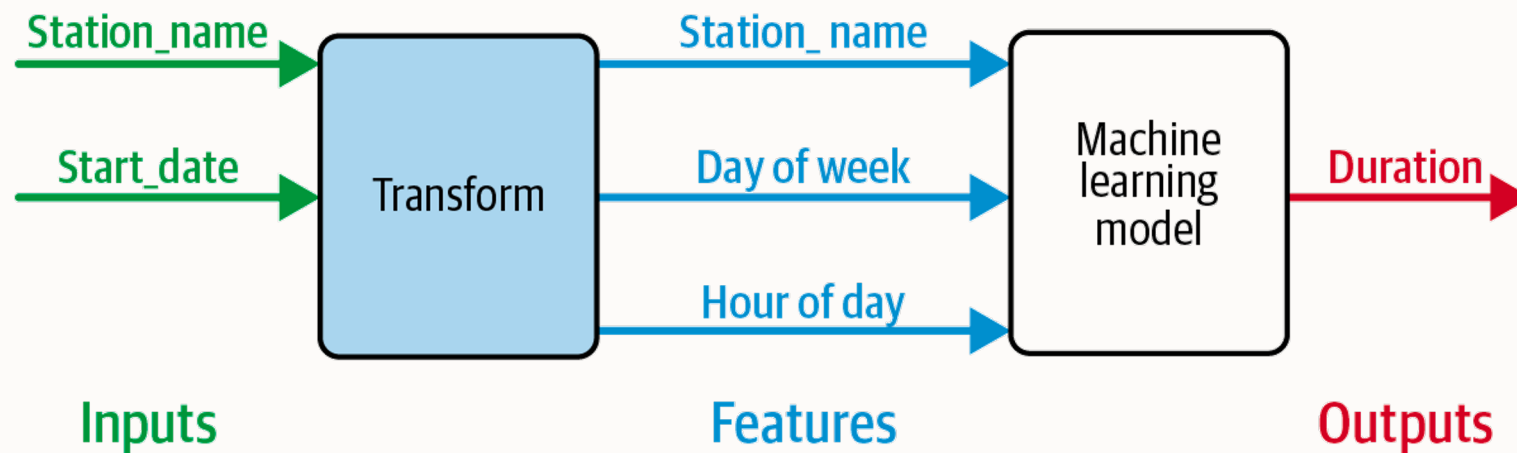
Design Pattern: Feature Store (1)

MLOps DP & BP (10)



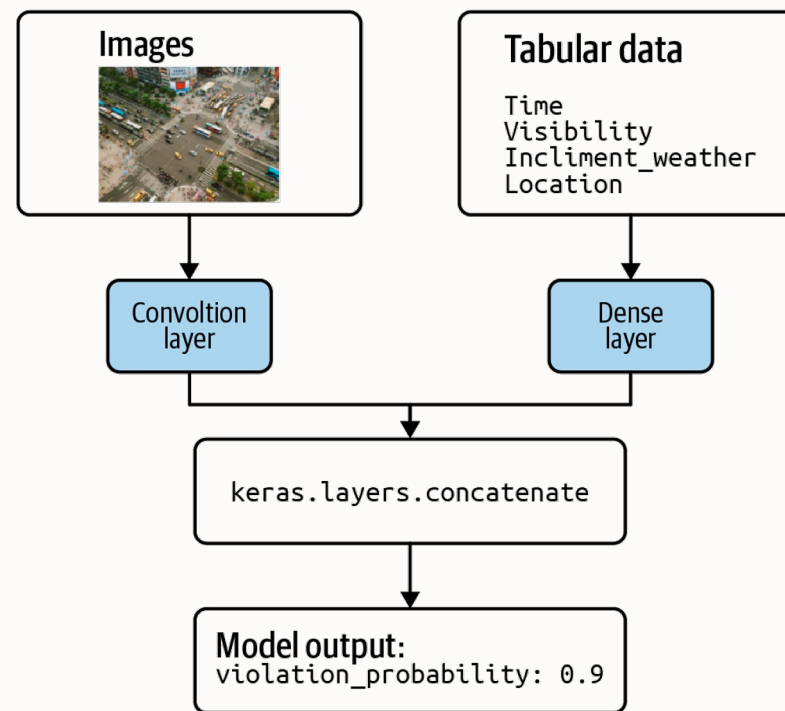
Design Pattern: Feature Store (2)

MLOps DP & BP (11)



Design Pattern: Transform

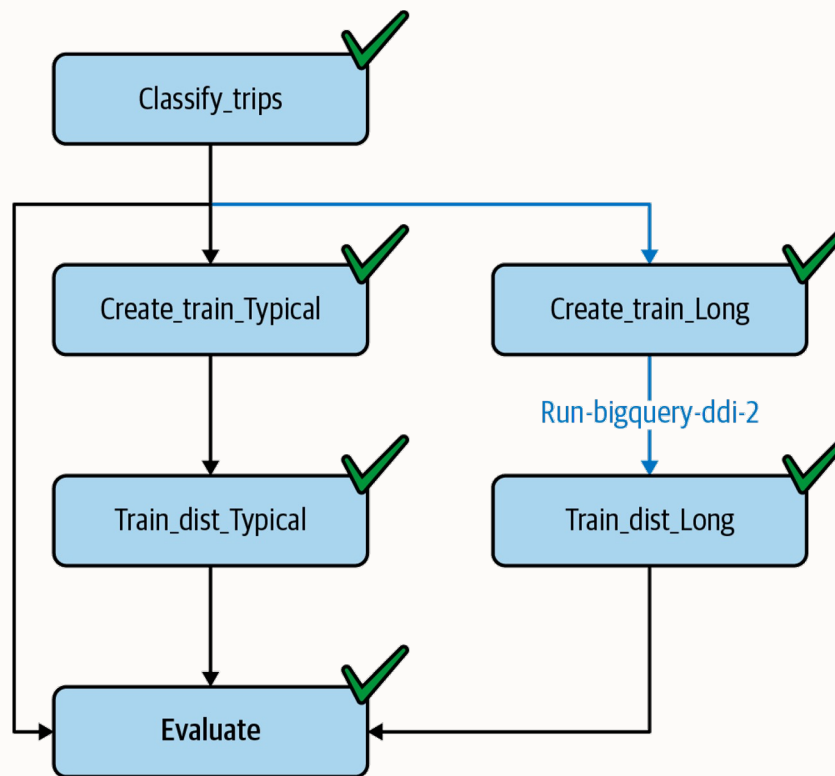
MLOps DP & BP (12)



Design Pattern: Multimodal Input

MLOps DP & BP (13)

		Predicted labels	
		Typical	Long
Actual labels	Typical	73.42%	26.58%
	Long	18.67%	81.33%

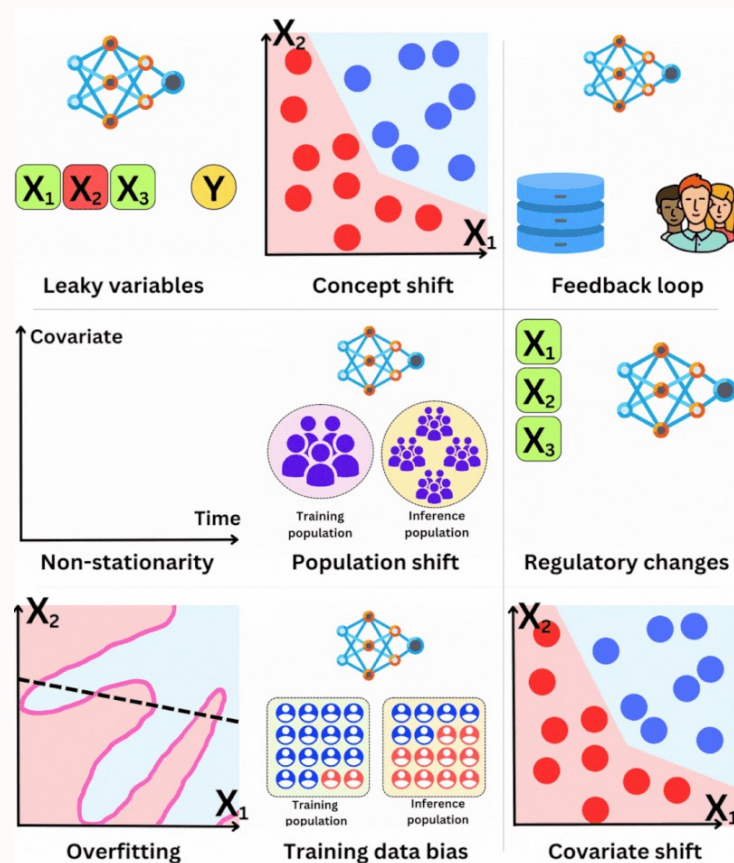


Design Pattern: Cascade

MLOps DP & BP (14)

- Les Bonnes Pratiques (Best Practices: BP) :
 1. S'assurer de la qualité et de la consistance des données.
 2. Utiliser les systèmes de contrôle de versions pour les données et les modèles.
 3. Utiliser des processus automatisés collaboratives et reproductibles.
 4. Utiliser des procédures de test et de validation automatisés.
 5. Utiliser des systèmes de surveillance, de contrôle et d'alerte.
 6. Adhérer à des conventions de nommages, de structuration de projet et de codage et vérifier régulièrement la qualité de ce dernier.

MLOps DP & BP (15)



Best Practice: What's going wrong?

MLOps DP & BP (16)

Best Practices for MLOps Lifecycle Management					
Data Preparation	Model Training	Model Evaluation	Deployment	Monitoring & Maintenance	Retiring & Replacing Models
					
• Ensure Data Quality • Feature Engineering • Data Versioning	• Experiment Tracking • Hyper-parameter Tuning • Cross-validation	• Choose Appropriate Evaluation Metrics • Use Multiple Evaluation Strategies • Consider Fairness & Bias	• Containerization • Automated Deployment • Version Control	• Set up Monitoring Infrastructure • Establish Alerting Mechanisms • Scheduled Retraining	• Model Performance Evaluation • Incremental Deployment • Document & Communicate

Best Practice: through the lifecycle.